

DeepMT Full Stack System Architecture

Platform Engineering, AI Orchestration & Network Deployment

1. Infrastructure: Mac Mini Local Deployment

Deploying DeepMT on a local Mac Mini requires a robust network and firewall setup to ensure secure inbound traffic and continuous uptime.

- **Process Management:** Use PM2 or macOS native launchd to ensure the backend Node.js/Python servers and DB run as background services on system boot.
- **Firewall Setup (pf/macOS Firewall):** Add rules to block unauthorized inbound connections. Only expose ports 80 (HTTP) and 443 (HTTPS) publicly.
- **Reverse Proxy & Tunneling:** Instead of messy home router port-forwarding (NAT), use **Cloudflare Tunnels (cloudflared)** or **Tailscale Funnel**. This exposes the Mac Mini server to the internet securely without opening physical router ports, instantly providing SSL/HTTPS.

2. Database & Authentication Layer

A relational database (PostgreSQL via Docker or a managed SQLite via Prisma/Drizzle) will handle states.

- **User Authentication:** Implement secure JWT (JSON Web Tokens) with standard Registration and Login flows. Passwords must be hashed using bcrypt or Argon2.
- **Data Schema:**
 - *Users Table:* id, email, password_hash, created_at, active_status
 - *Sessions Table:* id, user_id, title (auto-generated from first prompt), created_at
 - *Messages Table:* id, session_id, role (user/ai), content, created_at

3. AI Backend Engine (Turbo-Fieldfare)

The backend orchestration connects the database interface with the AI generation engine.

- **Engine Source:** Utilize drumih/turbo-fieldfare as the underlying AI response engine.
- **Streaming (SSE):** Instead of waiting for full generation, wrap the engine's output in a Server-Sent Events (SSE) or WebSocket stream. This streams raw tokens to the client in real-time.

4. Frontend UI/UX Flow

Built on the DeepMT Brand System (#F1F4E0 Light Canvas, #1C1C1C Dark Mode, #88B0A8 Brand Teal).

The Homepage: Features a sleek hero section using Playfair Display for the main header. The layout features scroll-triggered fade-ins and an interactive demo preview. A prominent '#88B0A8' accented Call-to-Action guides the user to 'Login / Register'.

The Auth Gateway: A minimalist, centered modal overlay holding the login/registration form with clean, sans-serif 1px border inputs. Focus states glow in Brand Teal.

The Chat Interface: Inspired by high-end modern editors. Large empty state featuring a watermark of the DeepMT flower, a sticky input box at the bottom, and clear, edge-to-edge message threads.

5. Advanced Signature Animations

A. The 'Thinking' Flower Rotation

When the AI is generating a response (waiting for the first token), the 5-petal flower favicon transforms into a custom loading state. The animation curve uses CSS custom cubic-bezier properties.

Animation Logic: Rotate standard 45 degrees, slow down (easing), and execute a rapid 360-degree spin before settling back to start.

```
@keyframes thinking-flower {
  0% { transform: rotate(0deg); }
  30% { transform: rotate(90deg); animation-timing-function: ease-out; }
  50% { transform: rotate(110deg); animation-timing-function: ease-in-out; }
  85% { transform: rotate(450deg); animation-timing-function: ease-in; }
  100% { transform: rotate(360deg); }
}
```

B. The Terminal Cursor Typing Indicator

While the response streams in, standard three-dot loaders are replaced by a sleek PC/Mac OS style blinking vertical cursor (|) appended to the end of the text node.

```
@keyframes blink-cursor {
  0%, 100% { border-right-color: transparent; }
  50% { border-right-color: #88B0A8; } /* Blinks in Brand Teal */
}
.typing-indicator::after {
  content: '|';
  border-right: 2px solid #88B0A8;
  animation: blink-cursor 1s step-end infinite;
  margin-left: 2px;
}
```